

Name:

Date:

Period:

Vector Addition with Vector Components

1. An airplane is taking off at a 57° angle with a velocity of 45 m/s. How fast is it traveling across the ground, as well as into the air?
2. Sandy Squirrel travels 18 m east and 41 m north to get to her secret stash of Emerald Nuts. Chester Chipmunk decides to take a short cut and travel in a straight line to the location of the (not so) secret stash. How far did he travel and at what angle?
3. Draw each of the following vectors and calculate the x - and y - components.

$$\vec{A} = 9 \text{ m @ } 35^\circ \text{ N of E}$$

$$\vec{B} = 12 \text{ m @ } 70^\circ \text{ E of S}$$

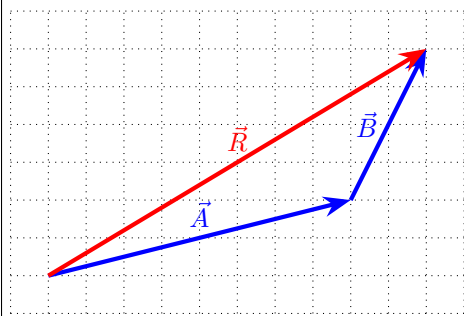
$$\vec{C} = 25 \text{ m @ } 12^\circ \text{ W of S}$$

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Vector Addition



4. Consider the following vectors:

$$\vec{D} = 30 \text{ m @ } 60^\circ \text{ N of W}$$

$$\vec{E} = 5 \text{ m, EAST}$$

Calculate $\vec{R} = \vec{A} + \vec{B}$.